

DISTANCE-TARGET IDENTIFICATION AND MULTI-RETURN SELECTION EMBODIMENTS

In various embodiments, the intraluminal compliance and distance profiling device incorporates computational logic configured to identify a distance target within an intraluminal environment and to select a reliable distance value in the presence of multiple sensor returns, variable reflectivity, motion artifacts, or anatomical variability.

Distance sensing modalities such as optical time-of-flight, LiDAR, or ultrasonic ranging may produce multiple candidate returns corresponding to different internal boundaries, fluid layers, tissue surfaces, window reflections, or partial occlusions. The computational system may therefore implement multi-return selection algorithms configured to choose a preferred return representing an intended measurement target.

In certain embodiments, the intended measurement target corresponds to a distal boundary such as a cervix-adjacent region, fornix-adjacent region, or other internal terminus. In other embodiments, the intended target is defined as a maximum stable distance return, a most consistent boundary over time, or a boundary meeting predefined confidence criteria, thereby enabling use without explicit anatomical labeling.

Multi-return selection may include filtering based on return strength, temporal stability, variance over repeated samples, consistency across multiple measurement angles, or agreement with learned baseline conditions. Outlier rejection and confidence weighting may be applied to remove transient or spurious returns.

In some implementations, the device acquires multiple samples across time or across small controlled positional changes and selects a final distance based on a robust statistic such as a median, trimmed mean, or best-of-N stability criterion. Sampling may include micro-oscillation of the device, user-guided micro-adjustment, or algorithmically timed capture windows during low-motion periods.

The system may incorporate sensor fusion to improve target identification. For example, motion sensing may identify stable windows for measurement, pressure signatures may confirm tissue contact conditions, and temperature sensing may confirm in-vivo conditions. Multi-factor fusion may increase confidence that the selected return represents valid intraluminal measurement.

When a reliable distance target is not detected, the device or associated software may provide corrective guidance. Guidance may include prompts to reposition, adjust angle, reduce motion, repeat sampling, or switch operating mode. In certain embodiments, the system records an uncertainty indicator rather than an absolute value when confidence thresholds are not achieved.

In certain embodiments, distance outputs may be transformed into normalized indices, categorical bands, or abstracted representations to preserve privacy and to maintain comparability across sensing modalities and target definitions. Normalization may include mapping absolute distances into standardized scales or ratios relative to baseline values.

Accordingly, distance-target identification and multi-return selection embodiments improve robustness of intraluminal ranging, reduce sensitivity to environmental variability, and prevent design-around implementations that rely on alternative target definitions or return-selection strategies while achieving equivalent measurement outcomes.